

CAN-BASED VEHICLE SAFETY & MONITORING SYSTEM

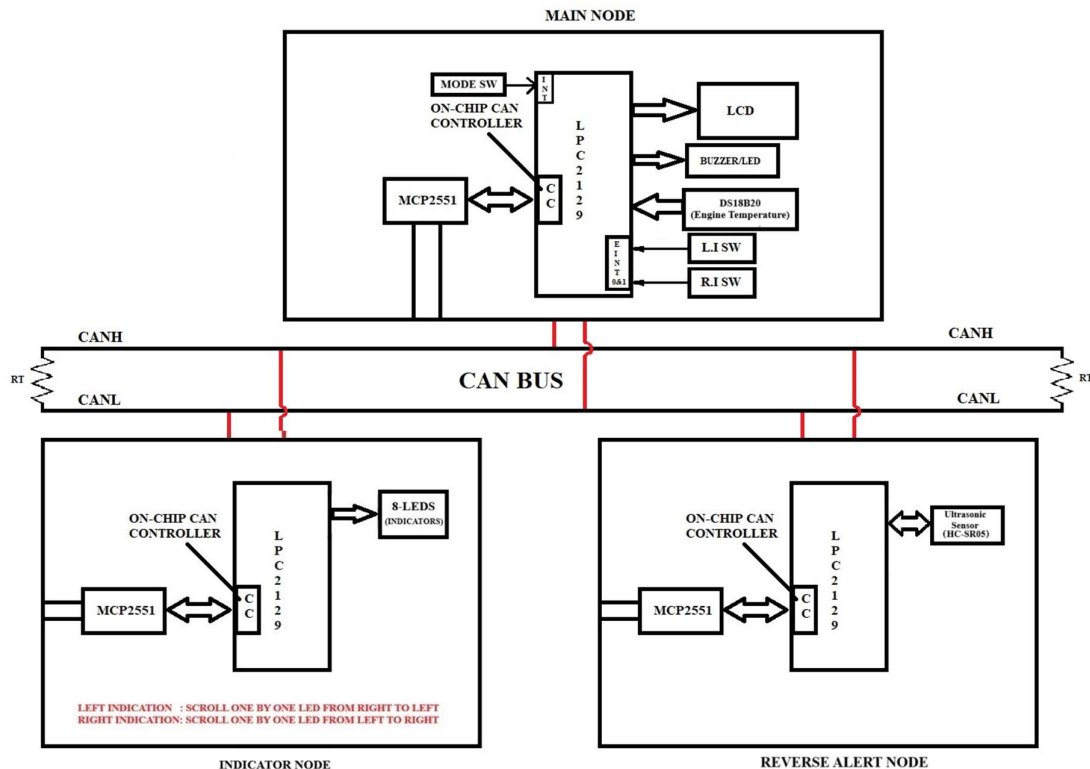
AIM:

To design and develop an automotive safety and monitoring system using the **Controller Area Network (CAN)** protocol, where a central (main) node monitors engine temperature, controls vehicle indicators, and processes reverse sensor data to provide real-time safety alerts through coordinated communication with multiple nodes.

INSIGHT:

- Knowledge of Embedded-C programming
- Thorough with the LPC2129 Architecture, General purpose I/O Interface, ADC, Interrupts & CAN interface
- Understanding of CAN protocol

BLOCK DIAGRAM:



HARDWARE REQUIREMENTS:

- LPC 2129
- CAN Transceiver (MCP2551)
- LEDS
- LCD
- Ultrasonic (HC-SR05)
- Switches
- DS18B20 Temperature Sensor
- USB to UART Converter

SOFTWARE REQUIREMENTS:

- 1) EMBEDDED C – PROGRAMMING
- 2) KEIL-C COMPILER
- 3) FLASH MAGIC

SEQUENCE TO BE FOLLOWED FOR IMPLEMENTATION:

- Create New Folder in your server/laptop/PC and save that folder with your project name
- Individually can check each module.
- First check lcd to display character constant, string constant and integer constant.
- Then develop logic for reading the DS18B20 (Engine Temperature) and display it on LCD.
- Then check the external interrupts code. Based on number of time interrupt is raised need to increase the count value and that should be displayed on LCD. (refer LMS for single & multiple interrupts sample codes)
- Then check the ultrasonic sensor working by reading the distance from the object and display it on LCD.
- Then download the CAN basic code from LMS and test it on hardware and analyse the working of that code.
- Once all the above modules are checked then start preparing the main code for each node.

MAIN NODE: In a continuous loop, the system reads the engine temperature using the DS18B20 sensor and displays it on the LCD. Whenever an interrupt is generated through external switches (SW1/SW2), the main node sends the appropriate indicator control signal to the Indicator Node via CAN communication. A mode switch is used to select the operating

mode. Initially, the vehicle operates in **forward mode**. When the mode switch is pressed, the system transitions to **reverse mode**. When the mode switch is pressed again, the system switches back to **forward mode**. In reverse mode, the system receives data from the reverse alert node. Based on this input, it provides an alert to the user by activating a buzzer or LED whenever necessary.

- **INDICATOR NODE:** Continuously waiting for the data from the main NODE using CAN. If any data is received from main NODE, based on that it is controlling the indicator signals connected to microcontroller. Refer the block diagram for how to give the indicator signals with the help of LED'S.
- **REVERSE ALERT NODE:** Continuously reading the HC-SR05 sensor information and set one limit value to the sensor reading. If the sensor value is below the limit value sends logic 1 to the main node using CAN protocol. Or if the sensor value is greater than the limit value then sends the logic 0 to the main node using CAN protocol.
- If you're getting this output, then your project is completed.

***** ALL THE BEST *****